Aror University of Art, Architecture, Design & Heritage Sukkur.

Department of Artificial Intelligence, Multimedia Gaming and Cyber Security

Object Oriented Programming(Fall-2025)

LAB No. 4

Objective of Lab No. 4:
After performing lab3, students will be able to:

- Implement Logical and Relational Operators in Java
- Implement Selection Statements in Java
- Implement Iteration Statements in Java
- Implement jump statements in java
- Implement Nested Loops in java
- Use All the above concepts to solve real life problems

Practice:

Task-01: Create a Java class and do the following:

a. Create an array to store 7 days of the week**.**
b. Use a foreach loop to print each day of the week.
c. Use a for loop to iterate over the array elements, and inside the loop use an if-else-if statement to check the day and print the class schedule for that day.

Solution:

```
public class WeekSchedule {

    public static void main(String[] args) {

        // a. Array to store days of the week
        String[] days = {"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday",
"Sunday"};
```

```
        // b. Using foreach loop to print days
        System.out.println("Days of the Week:");
        for (String day : days) {
            System.out.println(day);
        }

        // c. Using for loop with if-else-if
        System.out.println("\nClass Schedule:");
        for (int i = 0; i < days.length; i++) {

            if (days[i].equals("Monday")) {
                System.out.println("Monday: AI Programming Class");
            }
            else if (days[i].equals("Wednesday")) {
                System.out.println("Wednesday: Data Structures Lab");
            }
            else if (days[i].equals("Friday")) {
                System.out.println("Friday: Applied Physics");
            }
            else {
                System.out.println(days[i] + ": No Class Scheduled");
            }
        }
    }
}
```

Task-2: Generate the following sequences using both `for` loop and `while` loop separately.

```
2     40
5     35
8     30
11    25
14    20
17    15
20    10
23     5
```

i. The first column starts from **2** and increases by **3**.
ii. The second column starts from **40** and decreases by **5**.
iii. Print both columns side by side.
iv. Generate the sequence twice:

a. Using a for loop
b. Using a while loop

Solution:

```
public class SequenceGenerator {

    public static void main(String[] args) {

        System.out.println("Using for loop:");
        // Using for loop
        for (int i = 1, j = 39; i <= 22 && j >= 4; i += 3, j -= 5) {
            System.out.println(i + "     " + j);
        }
        System.out.println("\nUsing while loop:");
        // Using while loop
        int x = 1;
        int y = 39;
        while (x <= 22 && y >= 4) {
            System.out.println(x + "     " + y);
            x += 3;
            y -= 5;
        }
    }
}
```

Task-3: Create a matrix like shown below:

14	9	20	7
11	16	25	18
22	13	30	19

a. Divide each even number in the matrix by 2 and store the updated value back in the matrix.

b. Now use a for-each loop to iterate over the matrix and display only the odd numbers from the updated matrix.

c. Use a for loop to calculate and display the sum of the updated even numbers from the matrix.

Solution:

```
public class MatrixTask {

    public static void main(String[] args) {

        // Create the matrix

        int[][] matrix = {

            {14, 9, 20, 7},

            {11, 16, 25, 18},

            {22, 13, 30, 19}

        };

        // a. Divide each even number by 2 and update the matrix

        for (int i = 0; i < matrix.length; i++) {

            for (int j = 0; j < matrix[i].length; j++) {

                if (matrix[i][j] % 2 == 0) {

                    matrix[i][j] = matrix[i][j] / 2;

                }

            }
```

```
}
// b. Use for-each loop to display odd numbers from updated matrix
System.out.println("Odd numbers from updated matrix:");
for (int[] row : matrix) {
    for (int value : row) {
        if (value % 2 != 0) {
            System.out.print(value + " ");
        }
    }
}
// c. Use for loop to calculate sum of updated even numbers
int sum = 0;
for (int i = 0; i < matrix.length; i++) {
    for (int j = 0; j < matrix[i].length; j++) {
        if (matrix[i][j] % 2 == 0) {
            sum += matrix[i][j];
        }
    }
}
```

```
        System.out.println("\n\nSum of updated even numbers: " + sum);
    }
}
```

Task-4: Write a Java program to do the following:

a. Create an array of size 10 and populate it using a `do-while` loop with the following sequence:
`2, 5, 10, 17, 26, 37, 50, 65, 82, 0`

b. Now use a **`while`** loop to calculate the sum of even numbers from the array and skip the odd numbers using an appropriate jump statement.

c. Break the loop as soon as the value 65 is encountered.

Solution:

```
public class ArrayTask {

    public static void main(String[] args) {

        int[] arr = new int[10];

        // a. Populate array using do-while loop
        int i = 0;
        int n = 1;

        do {
            if (i == arr.length - 1) {
                arr[i] = 0;   // last element is 0
            } else {
                arr[i] = (n * n) + 1;  // n^2 + 1 sequence
            }
            n++;
            i++;
        } while (i < arr.length);
```

```
        // Display the array
        System.out.println("Array Elements:");
        for (int value : arr) {
            System.out.print(value + " ");
        }

        // b. Sum of even numbers using while loop
        // c. Break when 65 is encountered
        int sum = 0;
        int index = 0;

        while (index < arr.length) {

            if (arr[index] == 65) {
                break;  // stop loop when 65 is found
            }

            if (arr[index] % 2 != 0) {
                index++;
                continue;  // skip odd numbers
            }

            sum += arr[index];
            index++;
        }

        System.out.println("\n\nSum of even numbers (before 65): " + sum);
    }
}
```

Task-5: Design a simple Parking Slot Reservation System

The algorithm for the program is given below:

a. Create a parking structure by determining the number of floors and number of slots per floor. Floors and slots will determine the total number of parking spaces.

b. Use a do-while loop to display the following menu.
The loop will terminate when the user presses **3**:

i. Display available parking slots: **1**
ii. Reserve a parking slot: **2**
iii. Exit: **3**
iv. Enter your choice:

c. Use a switch-case statement inside the `do-while` loop:

i. If the user presses **1**, display the available parking slots.
Available slots are represented by false, and reserved slots are represented by true.

ii. If the user presses **2**, ask the user to enter the floor number and slot number
(Hint: subtract **1**, indexing starts from zero).

iii. Check whether the parking slot is already reserved or not.
If it is not reserved, reserve the slot; otherwise, display "Slot already reserved"**.**

iv. If the entered floor number or slot number is out of range, display "Invalid range"**.**

d. The default case of the switch should display "Invalid choice**"** if the user enters a number other than **1–3**.

Solution:

```
import java.util.Scanner;

public class ParkingReservation {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // a. Create parking structure

        System.out.print("Enter number of floors: ");
```

```
int floors = sc.nextInt();

System.out.print("Enter number of slots per floor: ");

int slots = sc.nextInt();

// false = available, true = reserved

boolean[][] parking = new boolean[floors][slots];

int choice;

// b. Menu using do-while loop

do {

   System.out.println("\n--- Parking Reservation Menu ---");

   System.out.println("1. Display available parking slots");

   System.out.println("2. Reserve a parking slot");

   System.out.println("3. Exit");

   System.out.print("Enter your choice: ");

   choice = sc.nextInt();

   // c. Switch-case inside do-while

   switch (choice) {

      case 1:

         // Display parking slots

         System.out.println("\nParking Slot Status:");
```

```
        for (int i = 0; i < floors; i++) {
            for (int j = 0; j < slots; j++) {
                System.out.print(parking[i][j] + " ");
            }
            System.out.println();
        }
        break;
    case 2:
        // Reserve a slot
        System.out.print("Enter floor number: ");
        int floor = sc.nextInt() - 1;
        System.out.print("Enter slot number: ");
        int slot = sc.nextInt() - 1;
        // Range checking
        if (floor < 0 || floor >= floors || slot < 0 || slot >= slots) {
            System.out.println("Invalid range.");
        }
        else if (parking[floor][slot]) {
            System.out.println("Slot already reserved.");
```

```
            }
            else {
                parking[floor][slot] = true;
                System.out.println("Parking slot reserved successfully.");
            }
            break;
        case 3:
            System.out.println("Exiting system...");
            break;
        default:
            // d. Invalid choice
            System.out.println("Invalid choice. Please enter 1 to 3.");
      }
    } while (choice != 3);
    sc.close();
  }
}
```

Aror University of Art, Architecture, Design & Heritage Sukkur.

Lab Exercises:

1. **Create a java class and do the following:**
    a. Create an array to store 4 seasons of an year
    b. Use foreach loop to print each of the array element
    c. Use for loop to iterate over the array elements, and inside the loop use if-else-if statement to check the season and print months of that particular season.

2. **Generate the following sequences using both for loop and while loop separately.**

```
run:
1       39
4       34
7       29
10      24
13      19
16      14
19      9
22      4
```

3. **Create a matrix like shown below:**

12	13	15	16
11	110	121	17
17	18	100	21

    a. Divide each even number from the matrix by 2 and store the updated value in the matrix.
    b. Now use for-each loop to iterate over the matrix and display the Odd Numbers from the matrix.
    c. Use for loop to do the sum of updated even numbers from the matrix

4. **Write a java program to do the following:**
    a. Populate an array having size 10 using a do while loop with following sequence: 1, 4, 9, 16, 25, 36, 49, 64, 81, 0 using a do while loop.
    b. Now use while loop to do the sum of odd numbers from this array and skip the even numbers using appropriate jump statements.
    c. Break the loop as soon as 81 is encountered.

5. **Design a simple seat reservation system for a theatre, the Algorithm for the program is given below:**
    a. Create theatre structure by determining the number of rows and number of columns, rows and columns will determine the total number of seats.
    b. Use a do while loop to display the following messages, loop will be terminated when user presses 3:

   i. Display available seats: 1
   ii. Reserve a seat: 2
   iii. Exit: 3
   iv. Enter your choice:

c. Now use a switch case statement inside the do while loop
   i. If user presses 1 then display the available seats, available seats are represented by false and unavailable seats are represented by true.
   ii. If user presses two, then ask the user row number and column number (Hint: subtract -1, indexing starts from zero)
   iii. Check if the seat is not reserved already then reserve the seat, otherwise display seat is already reserved.
   iv. If row number and column number are out of the range then display invalid range.

d. Default choice of switch will display invalid choice, if the user provides any number which is out of the given range (1-3)

6. Write java program to check wether a string is Palindrome or not (Hint: use string.charAt() method)

.